Tecnológico de Monterrey

Integración de seguridad informática en redes y sistemas de software (Gpo 501)

Activity
Stage 1 (Mobile Development) (Paco)

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13/10/2026

Who: What is the audience of our application, if you can define a few personas with characteristics, like age/profession among some of the factors we saw in class, that will help you identify NFRs.

Rescue coordinator / Dispatch manager [35 - 40 years]

- **Profession:** Operations coordinator at *Mas Bosque Manu* or lead paramedic officer
- **Tech literacy:** Moderate, uses desktop web interfaces or tablets to view reports and make resource allocation decisions.
- **Context and Needs:** Analyses historical incident trends, compares risk metrics across periods, and dispatches field support to high priority zones in Bosque La Primavera
- **Impact on NFRs:** Requires fast map rendering, intuitive filtering, offline / low-bandwidth report exports, and high availability during peak visitor hours.

Field Responder / Paramedic Volunteer [20-35 years]

- **Profession:** Volunteer paramedic university student volunteer, or trail rescue team member
- **Tech literacy:** High with mobile devices, often uses smartphones on the go in low connectivity areas.
- **Context and Needs:** Needs to view local terrain and quickly log incident locations on site or right after a rescue.
- **Impact on NFRs:** Requires mobile responsiveness, offline first data capture, low latency, and simplified UI suitable for outdoor/high glare environments.

Data entry and Admin operator [22 - 40 years]

- **Profession:** Administrative staff or designated data manager.
- **Tech Literacy:** Moderate, comfortable with web forms, database spreadsheets and basic mapping tools
- **Context and Needs:** Responsible for registering new incidents and maintaining historical database accuracy
- **Impact on NFRs:** requires strict data validation, secure authentication, and access control.

What: What are the main functional requirements for your mobile app? You can define the major functional requirements.

- **A very important component here, would be to identify the potential back-end services you will need, such as storage and potential back-end business logic.**

Core functional requirements

- **Interactive GIS Map:** Render base layer showing trails, access routes, water bodies, and public zones within Bosque La Primavera. Display georeferenced incident pins categorized by type, date, and severity level.
- **Incident Capture and Registration:** Simple entry form for logging incident details, such as location via map coordinates/trail, kilometer, date, severity and type.
- **Temporal Tracking and Filtering:** Multi-criteria filtering and side-by-side or period over period comparison to identify critical hotspots.
- **Data Export and Reporting:** Capability to export filtered map views or reports and maintain reusable geospatial data backups.

- **Data Confidence Alerts:** Visual notification when query parameters or timeframes yield insufficient data for statistical accuracy.

Required Back-end services and infrastructure

- **Geospatial Database:** PostGIS or MongoDB Geospatial for indexing geospatial coordinates, polygon trails and incident locations.
- **REST/GraphQL API Layer:** Microservices or API endpoints to handle spatial queries, filter requests, authentication, and incident CRUD operations.
- **File and Asset Storage:** Cloud storage, (FIrebase or AWS S3) for storing offline map tiles, satellite imagery layers, and export report PDFs.
- **Authentication and Authorization Service:** OAuth 2.0 service to manage role-based access control to the system.
- **Sync and Queue Engine:** Service to handle queued incident uploads created while field workers are out of cellular coverage.

How: Identify the NFRs for your application, such as security, accessibility, user/transaction volumes, among others.

Security

- **Authentication and RBAC:** Role-based access control restricting incident creation, database updates, and export privileges to authorized personnel.
- **Data integrity and protection:** HTTPS/TLS encryption in transit and AES-256 encryption at rest to safeguard sensitive rescue log data.

Performance and Speed:

- **Incident Log Time:** New incidents must be registered and reflected on the live interactive map.
- **Map Render Latency:** Spatial queries and period comparisons must render within 5secs under normal load.

Usability and Accessibility:

- **Self-service operational target:** At least 90% of active users must be able to operate map filters and register incidents independently without technical assistance.
- **Field visibility:** High contrast UI mode suitable for outdoor mobile screen visibility under direct sunlight.

Reliability and Data confidence:

- **Geospatial precision:** Minimum 90% historical georeferencing accuracy.
- **Data volume alerting:** System alerts when data thresholds are too low to render meaningful analytical conclusions.

Availability and Scalability:

- **Uptime:** operational availability for real-time access during high-visitation periods
- **Concurrency:** Support concurrent requests from up to 50 active field operators and dispatchers without degradation.